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HUDSON AND THAMES | Machine Learning in Econometrics

Beyond Econometrics: A Roadmap towards Financial Machine Learning

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01 Roadmap from Econometrics to Machine Learning:

01.1	Goal Setting
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A Roadmap from Econometrics to ML

STEP	ECONOMETRICS	CAVEAT	ML SOLUTION
Goal setting	Variance adjudication (in-sample)	Biased when confounders are omitted; poor forecasting performance	Out-of-sample prediction: Upper-boundary, importance, causation, reductionist, retriever
Visualization	Time plots, scatter plots, pie charts, histograms, heat maps	Not well suited for high- dimensional datasets with complex associations	t-SNE, networks, geospatial, classification trees, treemaps, etc.
Outlier detection	Winsorizing, trimming, Chauvenet's criterion, Mahalanobis distance, Dixon's Q test, etc.	Heuristic approaches	Anomaly detection methods, RANSAC
Feature extraction	PCA	It misses non-linear interactions and label information; it imposes a change of basis	Kernel-PCA, LDA, biclustering
Regression	Algebraic models	High risk of model mis-specification (wrong functional form, missing interaction effects, wrong assumptions, etc.)	Neural networks, SVR, GA, regression trees, etc.
Classification	Logit, probit	Linear specification; categorical or ordinal regressors require dummy variables	RF, SVC, k-NN, etc.
Feature importance	<i>p</i> -values	It assumes that the model is correctly specified; under multicollinearity, results are biased and not robust	MDI, MDA per cluster, Shapley values
Model selection/ Overfitting prevention	Forward selection, backward elimination, bidirectional elimination	Selection bias; in-sample procedures; mis-specification	Specification consistent with causal graph. Regularization, ensemble methods, etc.
Model validation	Adjusted <i>R</i> -squared (in-sample), analysis of residuals (in-sample)	It disregards prediction error; it does not weight errors by confidence	Out-of-sample (cross- validated): Explained variance, accuracy, F1, cross-entropy



Goal Setting

A poorly designed ML study, where the questions and goals are not clearly stated at the onset, is likely to lead to false discoveries.

ECONOMETRICS

Thought experiments are used for:

- **hypothesis testing** and **causality identification**.

***‘ceteris paribus’* argument**

- Evaluate the **sensitivity of one variable** to changes in another while the effect of all other relevant variables is controlled.
- Many **advantages** when the specification is correct.

Correlation vs Causation

- Chen and Pearl (2013) conduct a critical examination of the treatment of causation in econometrics, concluding the most econometrics textbooks **mistake correlation with causation**.

ML SOLUTION

Designed for **out-of-sample** forecasting:

Existence

- ML algorithms can point to the existence of an **undiscovered theorem**.
- If the model can't **confidently predict** the target variable then it is unlikely for the theory of this variable will emerge.

Importance

- **Discovers** the **variables** that should be part of the economic theory.

Reductionist

- **Discovers groups** within the data through clustering whose differentiating properties can then be analyzed.

Retriever

- **Detect outliers** based on the complex structure found in the data even if the structure is not explained to us.

Outlier Detection

Datapoints that do not appear to be generated by the same process as the rest and have a strong influence on the fitted model.

ECONOMETRICS

- Traditionally uses a number of **ad hoc techniques** to correct for the presence of outliers.

Winsorizing & Trimming

- Setting a **cap and a floor** on the observations in a dataset, where any observation in excess of the cap/floor is reset to the value of the cap/floor in winsorizing and removed in trimming.

Limitations

- Methods rely on **heuristics** or make **assumptions** about the data.

ML SOLUTION

- Takes into account the **complex structure** of the data.
- Using that knowledge to derive the probability that a particular observation is **anomalous**.

Random sample consensus(RANSAC)

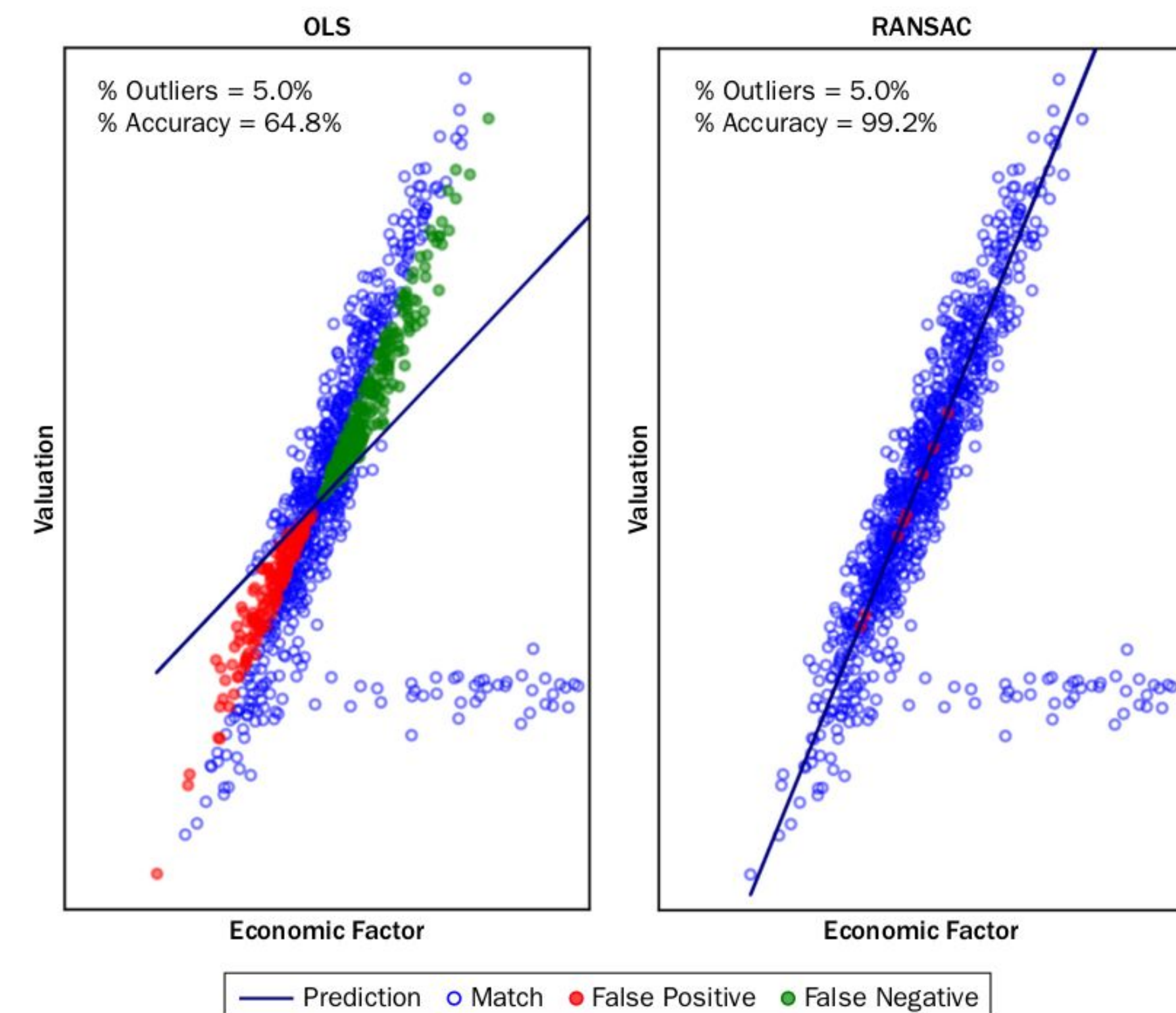
- This iterative method **splits the data set** between inliers and outliers such that the regression may be conducted on the inliers only.

Estimation in the Presence of Outliers: OLS vs. RANSAC

1 For cross-sectional studies, a **small number of outliers** can lead to the misclassification of a large number of securities.

2 The few outliers tilt the regression line, leading to many securities being falsely labeled as cheap (**false positives, in red**) or falsely labeled as expensive (**false negatives, in green**).

3 Outliers account for **5%** of the sample cause OLS to misclassify **35.2%** of the observations. In contrast, if we apply OLS excluding the outliers detected by RANSAC, the number of misclassified observations is only **0.8%** (mostly borderline cases).



Visualisation

Plotting the data helps to identify **outliers, missing information, or quality issues**.

ECONOMETRICS

The visual patterns that emerge from exploratory data analysis can suggest the **formulation of hypotheses**, which can then be tested quantitatively.

Time Plots

Box Plots

Bar Charts

Scatter Plots

Histograms

Distribution Plots

Pie Charts

Heat Maps

Pair Plots

ML SOLUTION

Datasets have grown in length, complexity, and dimensionality, which demand the use of **more modern representation methods**.

CART

t-SNE

Minimum spanning tree algorithm

Allows for the visualisation of **hierarchical relationships**.

Feature **Extraction**

Selecting from the set of features the minimum subset that allows us to achieve our goal.

ECONOMETRICS & ML SOLUTIONS

Principal Component Analysis(PCA)

Disadvantages:

1.1 Excludes **nonlinear interactions** between the features.

Solution: The kernel-PCA algorithm.

1.2 Extracts features with no knowledge of the **predicted variable**.

Solution: Linear discriminant analysis (LDA) method.

1.3 The resulting extracted features may not be **economically intuitive**.

Solution: Apply bi-clustering on the feature correlation matrix.

1.4 Eigenvectors associated with the smaller eigenvalues cannot be robustly estimated(present with low signal-to-noise ratio).

Solution: Apply de-noising and de-toning procedures.

Two problems are addressed:

1. The Curse of Dimensionality

2. Multicollinearity

Regression

ECONOMETRICS

Use an **algebraic specification** like a multivariate linear system of equations with interaction effects.

Algebraic functions cannot model complex data patterns like **nonlinear relations** that exhibit discontinuities or topological structures.

Multivariate Linear Regression

- High risk of **model misspecification**(wrong assumptions, wrong functional form).
- Unlike ML algorithms, econometric models **do not learn the structure** of the data via a specification search.

ML SOLUTION

Takes into account the **complex structure** of the data and test for **interaction effects** between variables.

Random Forest

Decision Trees

- This **divide-and-conquer algorithm** recursively partitions a dataset with complex patterns into subsets with simple patterns, which can then be fit independently with simple linear specifications.
- For this algorithm, **learning** means finding the splits that minimize the complexity of the subsets.

Classification

ECONOMETRICS

Most econometrics textbooks describe **two solutions** for classification problems:

Probit & Logit models

Disadvantages:

- Involves only **linear specifications**.
- Treatment of categorical (e.g., sectors of the economy) or ordinal (e.g., credit ratings) regressors requires encoding through **dummy variables**.
- The model may be biased by **imbalanced classes**.

ML SOLUTION

Solution to these disadvantages:

SMOTE and the near-miss algorithm

- Both SMOTE and near-miss rely on K-nearest neighbor(**KNN**), an ML algorithm.

Probabilistic classifiers

- The forecast has an **associated confidence**.
- Probabilistic classifiers include **naïve Bayes, decision trees, random forests, k-NN, ada-Boost, and neural networks**.

Feature Importance

Assess a feature's importance in explaining the target variable.

ECONOMETRICS

P-values

Assess a feature's importance through its **p-value**.

Limitations:

- 1 P-values rely on the **strong assumption** that the models is correctly specified.
- 2 When **multicollinearity** is present p-values are non-robust and often biased.

ML SOLUTION

Solution 1: Computational methods

MDI

- MDI's computational nature **avoids** the need for **distributional assumptions** that could be false.

MDA

- MDA method to discover the variables that should be part of the theory. A key difference that separates MDA from both p-value and MDI is that MDA **assesses significance out of sample**.

Shapely values

- Shapley values have several mathematical properties that make it **preferable over p-values or MDI**.

Solution 2: Dealing with substitution effects

- **Cluster** together interdependent features and derive MDI and MDA per cluster, rather than MDI and MDA per feature.

Model Selection / Overfitting prevention

A model that performs well in sample but performs poorly out-of-sample.

ECONOMETRICS

Favours the **most parsimonious model**.

1. Stepwise models

Forward selection

Backward elimination (avoid when multicollinearity is present)

Bidirectional elimination

Disadvantages:

- Decisions are based on **in-sample** statistics.
- Does not control for **selection bias** under multiple testing.

ML SOLUTION

1. Regularization

Ridge

LASSO

Elastic Net

- Hyper-parameters that control the penalization functions are found through cross-validation, hence **minimizing the model's error** on the test set.

2. Early stopping

3. Pruning (Used for decision trees)

4. Drop out (Used in neural network's hidden layers)

5. Kernel-based algorithms (For example: support vector machines)

6. Bagging

7. Causal graph

PC algorithm

Spirtes and Glymour (1991)

Model Validation

The discrepancy between expected and observed values.

ECONOMETRICS

Evaluates performance **in-sample**

Coefficient of Determination - R-squared & Adjusted R-squared

- **Testing the residuals** for potential violation of the model’s assumptions is as important as achieving a significant R2.
- Common **tests include:**

Jarque-Bera’s normality test

Durbin-Watson’s autocorrelation test

White’s heteroscedasticity test

ML SOLUTION

Evaluates performance **out-of-sample**

Cross Validation

Regression

Explained variance

Mean Absolute Error

Median Absolute Error

Binary Classification

Recall

Precision

F1-score

Multi-class Classification

Accuracy

Cross-entropy

Bet sizes

(model confidence)

THANK YOU FOR JOINING!

